FUNCTIONS & GRAPHS 2

Practice Bank with Full Solutions

Polynomial, Reciprocal & Rational Graphs

10 Core + 10 Challenge + 10 Advanced

Practice principle: read algebra → extract features → predict behaviour → solve / interpret → check against structure.

CORE

C1 - Read a cubic from its factors
Let $f(x) = (x+4)(x-1)(x-3)$. Without a value table: (a) state all x-intercepts; (b) find the y-intercept; (c) state whether the graph crosses or touches at each root; (d) describe the end behaviour.

C2 - Repeated root
Let $g(x) = -(x+2)^2(x-5)$. Determine the roots, identify the touching and crossing roots, find the y-intercept, and describe the end behaviour.

C3 - Quartic structure
Let $h(x) = -2(x-1)(x+3)(x-4)(x+1)$. State the four roots, crossing/touching behaviour, the y-intercept, and the end behaviour.

C4 - Sign from a cubic
Let $f(x) = x(x-2)(x+3)$. Without expanding, determine where $f(x) > 0$ and where $f(x) < 0$.

C5 - Reciprocal structure
For $f(x) = -3/x$, determine the domain, both asymptotes, the sign for $x < 0$ and $x > 0$, and the quadrants containing the graph.

C6 - Translated reciprocal
For $f(x) = 4/(x+2) + 1$, find the domain, vertical/horizontal asymptotes, x-intercept and y-intercept.

C7 - Simple rational graph
Let $f(x) = (x-4)/(x+1)$. State the domain; rewrite it as $A + B/(x+1)$; state both asymptotes; find both axis intercepts; determine where f is positive/negative.

C8 - Intersections as solutions
Find the intersection points of $y = x(x-2)(x+2)$ and $y = 5x$. Hence state the number of real solutions of $x(x-2)(x+2) = 5x$.

C9 - Polynomial inequality
Solve $(x+2)(x-1)(x-4) > 0$.

C10 - Transform reciprocal features
Let $g(x) = 2/(x-1) - 3$ and $h(x) = g(x+4) + 5$. Find the vertical asymptote, horizontal asymptote and x-intercept of h.

CHALLENGE

H1 - Reconstruct a cubic
A cubic crosses the x-axis at $x = -3$, touches at $x = 2$, and passes through $(3, -24)$. Find a cubic function satisfying these conditions and state its end behaviour.

H2 - Reconstruct a quartic
A quartic touches at $x = -2$, crosses at $x = 1$ and $x = 4$, and passes through $(0, -32)$. Find a quartic of least possible degree and determine its end behaviour.

H3 - Reconstruct a reciprocal graph
A reciprocal-type function has vertical asymptote $x = 3$, horizontal asymptote $y = 2$, and crosses the x-axis at $x = 1$. Find $f(x) = a/(x - 3) + 2$ and its y-intercept.

H4 - Rational comparison inequality
Solve $(2x + 1)/(x - 3) > 1$. Make clear why $x = 3$ is excluded.

H5 - Line-cubic intersections
Find all intersections of $y = (x - 1)(x + 1)(x - 3)$ and $y = x - 3$. Interpret the result graphically.

H6 - Number of solutions from turning levels
A positive-leading cubic has local maximum value 4 and local minimum value -3. State the number of distinct real solutions of $f(x) = k$ for: (a) $k > 4$; (b) $k = 4$; (c) $-3 < k < 4$; (d) $k = -3$; (e) $k < -3$.

H7 - Find the reciprocal parameter
Let $f(x) = a/(x + 1) - 2$ and suppose the graph passes through $(1, 1)$. Find a, both asymptotes and both axis intercepts.

H8 - Can the graphs intersect?
Determine whether $y = 6/(x + 2)$ and $y = 2 - x$ intersect. Justify algebraically and interpret graphically.

H9 - Reconstruct from mixed features
A reciprocal-type function has $f(x) = a/(x - 1) + k$, crosses the x-axis at $x = 4$, and crosses the y-axis at $(0, 4)$. Find a, k, the horizontal asymptote, and the intervals where $f(x) > 0$.

H10 - Transformed cubic roots
A cubic f has positive leading coefficient and three simple roots $-2, 1, 5$. Define $g(x) = -f(2x + 4)$. Find the roots of g, say whether each is crossing/touching, and state the end behaviour.

ADVANCED

A1 - Quartic reconstruction and inequality
A quartic touches at $x = -3$, crosses at $x = 1$ and $x = 4$, and passes through $(0, 72)$. Construct the least-degree polynomial, state end behaviour, and solve $f(x) < 0$.

A2 - Parameterised line-parabola intersections
Consider $y = x^2 - 2x + 5$ and the family $y = mx + 3$. Determine the values of m giving (a) two distinct real intersections; (b) exactly one; (c) none. Interpret the boundary cases graphically.

A3 - Reconstruct a rational function
A reciprocal-type function has vertical asymptote $x = 2$, horizontal asymptote $y = -1$, and passes through $(0, 3)$. Find the function, its x-intercept, and solve $f(x) > 0$.

A4 - Transform a rational graph
Let $f(x) = 3/(x - 2) - 1$ and $g(x) = -2f(3x + 6)$. Without a value table find the domain, both asymptotes, x-intercept and y-intercept of g.

A5 - Compare two rational functions
Solve $2/(x-1) > 1/(x+2)$. Show all critical values and exclude undefined values correctly.

A6 - Parameterised cubic-line intersections
Consider $f(x) = x^3 - 3x$ and lines $y = kx$. Determine k for which there are (a) three distinct real intersection points; (b) exactly one distinct real intersection. Explain the boundary.

A7 - Repeated-root reconstruction
A monic cubic has repeated root $x = 2$ and constant term -6. The third root is real. Find it, expand the cubic as $x^3 + ax^2 + bx - 6$, and describe behaviour at each root.

A8 - Transformed quartic structure
Let $f(x) = (x+2)^2(x-1)(x-4)$ and $g(x) = -f(2x-2)$. Without expanding: find all roots, identify the touching root, state end behaviour, and solve $g(x) > 0$.

A9 - Hole + asymptote + inequality
Let $r(x) = ((x-3)(x+1))/((x-3)(x-2))$. State the full domain, simplify for permitted x, identify the vertical/horizontal asymptotes and the hole, find the x-intercept, and solve $r(x) > 0$.

A10 - Reconstruct then solve
Let $f(x) = (ax+b)/(x-2)$. Its horizontal asymptote is $y = 3$ and its x-intercept is $x = -1$. Find a, b, rewrite in translated reciprocal form, state both asymptotes, and solve $f(x) > 6$.

FULL SOLUTIONS - CORE

C1
Roots: $-4, 1, 3$. $f(0) = 12$, so the y-intercept is $(0, 12)$. All roots are simple, so all are crossings. Leading term x^3: left down, right up.

C2
Roots -2 and 5. The squared root -2 is a touch; 5 is a crossing. $g(0) = -(2)^2(-5) = 20$. Leading term $-x^3$: left up, right down.

C3
Roots $-3, -1, 1, 4$, all simple crossings. $h(0) = -2(-1)(3)(-4)(1) = -24$. Leading term $-2x^4$: both ends down.

C4
Critical values $-3, 0, 2$, all simple. Signs alternate for this positive-leading cubic: $-, +, -, +$. Hence $f > 0$ on $(-3, 0) \cup (2, \infty)$ and $f < 0$ on $(-\infty, -3) \cup (0, 2)$.

C5
Domain $x \neq 0$. Vertical asymptote $x = 0$, horizontal asymptote $y = 0$. If $x < 0$, $-3/x > 0$; if $x > 0$, $-3/x < 0$. The graph lies in quadrants II and IV.

C6
Domain $x \neq -2$. Asymptotes $x = -2$ and $y = 1$. For the root, $4/(x + 2) = -1$, so $x = -6$. $f(0) = 3$, giving $(0, 3)$.

C7
Domain $x \neq -1$. Since $x - 4 = (x + 1) - 5$, $f(x) = 1 - 5/(x + 1)$. Asymptotes $x = -1, y = 1$. Root $x = 4$, y-intercept $(0, -4)$. Sign: positive for $x < -1$ or $x > 4$, negative for $-1 < x < 4$.

C8
Set equal: $x(x^2 - 4) = 5x$, so $x(x^2 - 9) = 0$. Thus $x = -3, 0, 3$. Using $y = 5x$, the intersections are $(-3, -15), (0, 0), (3, 15)$. There are three real solutions.

C9
Simple roots $-2, 1, 4$ split the positive-leading cubic into signs $-, +, -, +$. Thus $-2 < x < 1$ or $x > 4$.

C10
$h(x) = 2/(x + 3) + 2$. Hence asymptotes $x = -3, y = 2$. Setting $h = 0$ gives $2/(x + 3) = -2$, so $x = -4$.

FULL SOLUTIONS - CHALLENGE

H1
Crossing at -3 gives $(x + 3)$; touching at 2 gives $(x - 2)^2$. So $f = k(x + 3)(x - 2)^2$. Using $(3, -24)$: $-24 = 6k$, hence $k = -4$. Therefore $\boxed{f(x) = -4(x + 3)(x - 2)^2}$. Negative-leading cubic: left up, right down.

H2
$f = k(x + 2)^2(x - 1)(x - 4)$. Using $f(0) = -32$: $-32 = 16k$, so $k = -2$. Thus $\boxed{f = -2(x + 2)^2(x - 1)(x - 4)}$. Negative quartic: both ends down.

H3
Asymptotes give $f = a/(x - 3) + 2$. Root at 1: $0 = a/(-2) + 2$, so $a = 4$. Hence $f = 4/(x - 3) + 2$. Then $f(0) = 2/3$.

H4
$\frac{2x+1}{x-3} - 1 = \frac{x+4}{x-3} > 0$. Critical values $-4, 3$. Positive when numerator and denominator have the same sign: $\boxed{x < -4 \text{ or } x > 3}$. $x = 3$ is undefined.

H5
Set equal: $(x-1)(x+1)(x-3) = x-3$. Factor $(x-3)[(x-1)(x+1)-1] = 0$, so $(x-3)(x^2-2) = 0$. Hence $x = 3, \pm\sqrt{2}$. With $y = x-3$, intersections are $(3, 0)$, $(\sqrt{2}, \sqrt{2}-3)$, and $(-\sqrt{2}, -\sqrt{2}-3)$.

H6
A positive-leading cubic rises to max 4, falls to min -3, then rises. Thus numbers of distinct solutions: $k > 4$: 1; $k = 4$: 2; $-3 < k < 4$: 3; $k = -3$: 2; $k < -3$: 1.

H7
$1 = a/2 - 2$, so $a = 6$. Thus $f = 6/(x+1) - 2$. Asymptotes $x = -1, y = -2$. Root: $6/(x+1) = 2 \Rightarrow x = 2$. $f(0) = 4$.

H8
$6/(x+2) = 2 - x$ gives $6 = (2-x)(x+2) = 4 - x^2$, hence $x^2 = -2$. No real solution, so the graphs do not intersect.

H9
From the root $x = 4$: $0 = a/3 + k$, so $k = -a/3$. From $(0, 4)$: $4 = -a + k$. Hence $a = -3, k = 1$. Thus $f = -3/(x-1) + 1 = (x-4)/(x-1)$. Horizontal asymptote $y = 1$. Sign is positive for $x < 1$ or $x > 4$.

H10
Roots satisfy $2x + 4 = r$ for original roots $r = -2, 1, 5$. Thus new roots $-3, -3/2, 1/2$. All remain simple crossings. The outer minus sign turns the positive-leading cubic into a negative-leading cubic: left up, right down.

FULL SOLUTIONS - ADVANCED

A1
$f = k(x+3)^2(x-1)(x-4)$. Using $f(0) = 72$: $72 = 36k$, so $k = 2$. Thus $f = 2(x+3)^2(x-1)(x-4)$. Positive quartic: both ends up. The squared factor does not change sign, so $f < 0$ exactly on $\boxed{1 < x < 4}$.

A2
Intersections satisfy $x^2 - 2x + 5 = mx + 3$, i.e. $x^2 - (m+2)x + 2 = 0$. Discriminant $\Delta = (m+2)^2 - 8$. Two intersections when $\Delta > 0$: $\boxed{m < -2 - 2\sqrt{2} \text{ or } m > -2 + 2\sqrt{2}}$. One intersection when $\Delta = 0$: $\boxed{m = -2 \pm 2\sqrt{2}}$ (tangent). No intersection when $\boxed{-2 - 2\sqrt{2} < m < -2 + 2\sqrt{2}}$.

A3
Asymptotes give $f = a/(x-2) - 1$. Using $(0, 3)$: $3 = a/(-2) - 1$, so $a = -8$. Thus $f = -8/(x-2) - 1$. Root: $-8/(x-2) = 1 \Rightarrow x = -6$. Sign analysis around -6 and 2 gives $\boxed{-6 < x < 2}$.

A4
Vertical asymptote occurs when $3x + 6 = 2$, so $x = -4/3$ and the domain excludes it. Original horizontal asymptote is -1, so after multiplication by -2 it becomes $y = 2$. The original root is 5, so $3x + 6 = 5 \Rightarrow x = -1/3$. Finally $g(0) = -2f(6) = -2(3/4 - 1) = 1/2$.

A5
Combine: $\frac{2}{x-1} - \frac{1}{x+2} = \frac{x+5}{(x-1)(x+2)} > 0$. Critical values $-5, -2, 1$. Signs are $-, +, -, +$. Strict inequality and domain exclusions give $\boxed{(-5, -2) \cup (1, \infty)}$.

A6
Intersections satisfy $x^3 - 3x = kx$, so $x[x^2 - (k+3)] = 0$. $x = 0$ always. Two additional distinct real roots occur when $k + 3 > 0$, so $\boxed{k > -3}$ gives three intersections. If $k \leq -3$, no additional distinct real roots exist; exactly one distinct intersection remains. Boundary $k = -3$ gives the triple root $x = 0$.

A7
$f = (x-2)^2(x-r)$ because the cubic is monic. Constant term $(-2)^2(-r) = -4r = -6$, so $r = 3/2$. Expanding gives $\boxed{x^3 - \frac{11}{2}x^2 + 10x - 6}$. The graph crosses at $x = 3/2$ and touches at $x = 2$.

A8
Solve $2x - 2 = r$ for original roots $r = -2$ (double), 1, 4. New roots are 0 (double), 3/2, 3. The outer minus sign makes the quartic negative-leading, so both ends down. The double root does not change sign; the simple roots do. Hence $\boxed{g(x) > 0 \text{ for } 3/2 < x < 3}$.

A9
Original domain excludes 2 and 3. For permitted x, $r = (x+1)/(x-2)$. Thus vertical asymptote $x = 2$, horizontal asymptote $y = 1$, and the cancelled value gives a hole at $(3, 4)$. Root $x = -1$. The simplified sign is positive for $x < -1$ or $x > 2$, but $x = 3$ is still excluded. Hence $\boxed{(-\infty, -1) \cup (2, 3) \cup (3, \infty)}$.

A10
Horizontal asymptote $y = 3$ gives $a = 3$. Root $x = -1$ gives $-3 + b = 0$, so $b = 3$. Then $f = (3x+3)/(x-2) = 3 + 9/(x-2)$. Asymptotes $x = 2, y = 3$. Solve $f > 6$: $(3x+3)/(x-2) > 6 \iff (x-5)/(x-2) < 0$, hence $\boxed{2 < x < 5}$.

Hint ladders and interactive-generation rules

Polynomial structure. Identify factors → roots → multiplicities → leading term.

Reconstruction. Roots → factors; touching root → repeated factor; introduce k; use one extra point.

Reciprocal/rational. Find excluded inputs; determine asymptotes; then intercepts and signs.

Inequality. List zeros and excluded values; sort critical values; determine signs interval by interval.

Intersections. Set functions equal. Number of real algebraic solutions equals number of graph intersections.

Transformations. Transform roots/asymptotes/features, not a table of points.

Safe randomization: generate valid algebra first, then derive all roots, multiplicities, signs, asymptotes, holes and graph features. Never randomize algebra and rendered graph independently. For rational functions, always test common-factor cancellation before classifying a denominator zero as a vertical asymptote.

Common error. Anti-answer-leak rule: if a problem asks for roots, asymptotes, intersections, hole coordinates, sketch or inequality intervals, those targets must not already be labelled/shaded in the unsolved visual.

This Practice Bank contains original generated questions. No item is claimed to be an official Cambridge examination question. The difficulty design is calibrated to Cambridge secondary / Additional Mathematics foundations with Cambridge 9709 P1-style reasoning as the Advanced ceiling.